VIVEK RAJ

Applied Research Scientist · Real-Time Computer Vision

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SUMMARY

PhD candidate at IIT Delhi who builds real-time computer vision systems that turn camera images into continuous control signals, and validates them with real users. Built a markerless system that let a person with a transradial amputation drive a bionic hand from camera images alone, running on a laptop CPU. Work spans data collection, automatic labelling with foundation models, model training, CPU deployment, closed-loop testing and statistical evaluation.

EXPERIENCE

Doctoral Researcher | RISE Lab, Centre for Biomedical Engineering, IIT Delhi | Oct 2020 – present

Clinical partner: AIIMS New Delhi

- Built a markerless vision pipeline that drove an Aether Zeus bionic hand in real time. A MobileNetV2-FPN segmenter exported to ONNX feeds 11 engineered features to linear and SVR decoders on a laptop CPU. In online testing, a participant with a transradial amputation succeeded in 54 of 54 trials across three degrees of freedom.
- Generated segmentation labels without manual annotation: ran SAM 2.1 on dual cloud GPUs, filtered masks through four quality-control rounds, then distilled them into a 4.2 M-parameter student with a confidence-weighted BCE, Dice and boundary loss. After a label review it reached IoU 0.934 on held-out sessions and 0.906 on unseen participants (leave-one-subject-out).
- Quantized the segmenter to INT8 with static post-training quantization. The model shrank from 17.3 MB to 4.7 MB and ran 1.7 times faster on a laptop CPU (9.2 ms vs 15.7 ms per frame) with no loss of IoU. The full pipeline runs in about 20 ms per frame.
- Diagnosed a drop in closed-loop throughput between sessions by replaying recorded video through each combination of segmenter and feature set. The deployed configuration was the most accurate offline, which located the cause in closed-loop holding behaviour rather than in the model.
- Built a single-camera reflective-marker tracker (Canny edges, morphology, CLAHE, area and circularity filtering, ratio-test matching, constant-velocity Kalman filter) with translation-, rotation- and scale-invariant triplet-angle features. It runs online at 12.8 Hz (68.7 ms per frame) with ten markers and about 29 Hz with three. Across 16 users it achieved 90.6–100% task success in 13 non-disabled participants and 98.8% (160 of 162 trials) in 3 participants with transradial amputation.
- Built synchronised multi-sensor acquisition (USB camera, IMU over TCP, data glove) and released an open-source MATLAB toolbox for the xIMU3 IMU (400 Hz streaming).
- Ran and analysed ethics-approved human studies at IIT Delhi and AIIMS New Delhi. Published in *IEEE Sensors Journal*; two manuscripts under review or revision; one invention disclosure filed (FITT, IIT Delhi, 2024).

M.Tech. Researcher | NIT Silchar | 2018 – 2020

- Classified cognitive tasks from EEG with engineered time- and frequency-domain features and an SVM (98% cross-validated accuracy); published at IEEE ASPCON 2020.

SKILLS

Computer vision and ML: PyTorch · segmentation-models-pytorch · OpenCV · SAM 2.1 · ONNX Runtime · INT8 quantization · scikit-learn · semantic segmentation · object tracking · Kalman filtering · regression
Engineering: Python · MATLAB · real-time capture and control loops · UDP/TCP device interfaces · Git · cloud GPU training (Kaggle, vast.ai)
Evaluation: cross-validation design · closed-loop user testing · Fitts'-law throughput · repeated-measures ANOVA
Currently learning: C++, ROS 2, TensorRT

EDUCATION

Ph.D., Biomedical Engineering | IIT Delhi | Oct 2020 – present | CGPA 8.45/10
M.Tech., Instrumentation Engineering | NIT Silchar | 2020 | CGPA 9.0/10, Distinction
B.Tech., Electronics and Instrumentation Engineering | University of Kalyani | 2018 | CGPA 7.81/10

SELECTED PUBLICATIONS

1. V. Raj *et al.*, "Toward markerless, noncontact optical myography for prosthetic control of multiple degrees of freedom," *IEEE Sensors Journal*, 25(4), 7154–7162, 2025.
2. V. Raj *et al.*, "Optical myography-based measurement of proportional hand and wrist kinematics for prosthetic control," submitted, *Annals of Biomedical Engineering*, 2026.
3. V. Raj, B. Mukherjee, "Monocular optical myography for real-time proportional control of wrist and hand motions," under revision, *IEEE Sensors Journal*, 2026.